Relational Learning with Covariance Information

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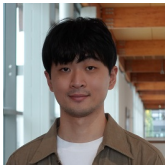
Acknowledgements

Andrea Cavallo

Mohammad Sabbaqi



Zhan Gao

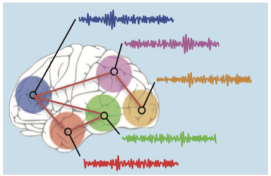
Madeline Navarro

Santiago Segarra

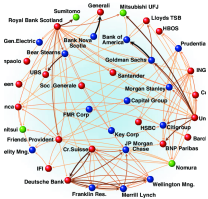
TU Delft | AI Initiative

Covariance relationships between data points

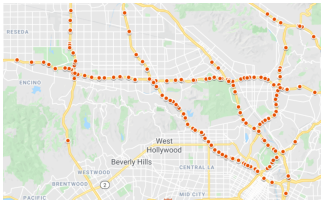
- Data usually contains hidden interconnections



Brain



Financial

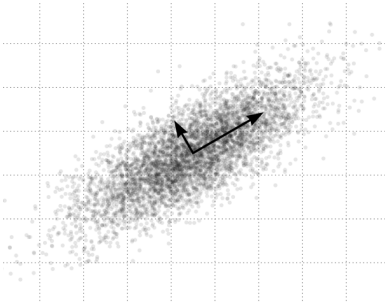


Traffic

- One way to capture these relations is through the covariance matrix
  $\Rightarrow \mathbf{C} = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\mathsf{T}]$

Principal Component Analysis (PCA)

- Project the data onto the covariance eigenspace
  - $\Rightarrow \mathbf{C} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$
  - $\Rightarrow \tilde{\mathbf{x}} = \mathbf{V}^\top \mathbf{x}$
- Select (filter) the directions that maximize the variance of data points
  - $\Rightarrow$ Used for dimensionality reduction by selecting only a few eigenvectors



Principal directions

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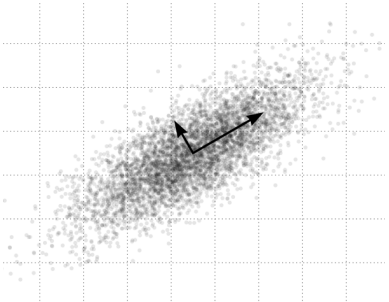
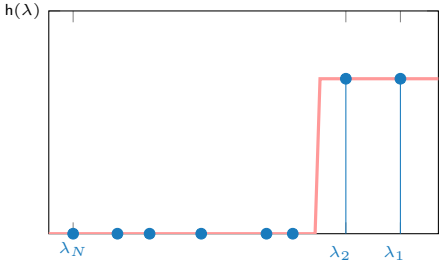


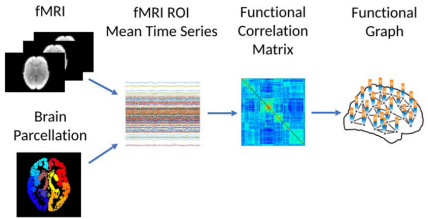
Principal directions



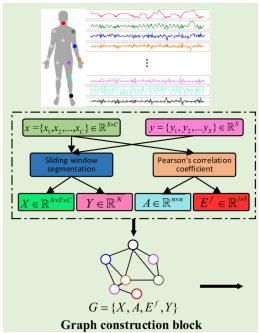
PCA filtering

Topology Inference

- Graphical Lasso, Collaborative Filtering, Graph Stationarity



Brain fMRI data



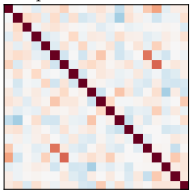
Human motion sensor recordings

- Use this topology for downstream processing

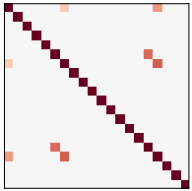
Wang et al., MhaGNN, IEEE Transactions on Instrumentation and Measurement, 2023
Li et al., BrainGNN: Interpretable brain graph neural network for fMRI analysis. Medical Image Analysis, 2021

- In practice, we work with estimates
  $\Rightarrow \hat{\mathbf{C}} = \frac{1}{t}\sum_{t'=1}^{t} \mathbf{x}_{t'}\mathbf{x}_{t'}^{\mathsf{T}}$
- May not reflect the true underlying structure
  $\Rightarrow$ Sample estimator $\hat{\mathbf{C}}$ is noisy
  $\Rightarrow$ When number of samples t is of same order of data dimension N

Empirical covariance

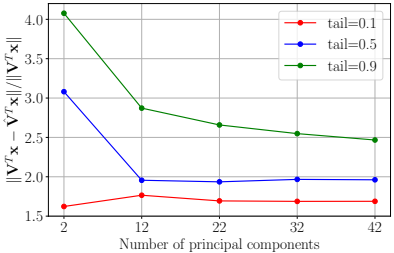


True covariance



Finite-data effect on PCA

- PCA is unstable to covariance estimation errors



Empirical stability

Johnstone & Lu. On consistency and sparsity for principal components analysis in high dimensions. JASA. 2009

Learning with covariances

- Finite-data: alterations in the principal directions
- Sparse estimates: alterations in the principal directions
- Solution: graph neural networks
  - $\Rightarrow$ Covariance-based neural networks improve stability
  - $\Rightarrow$ Can be made efficient through sparsification
  - $\Rightarrow$ Work well in a variety of settings: static, temporal, and biased data

Outline

- Covariance Neural Networks
- Sparse VNNs
- Spatiotemporal VNNs
- Fair VNNs
- Conclusions

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- **Covariance Neural Networks**
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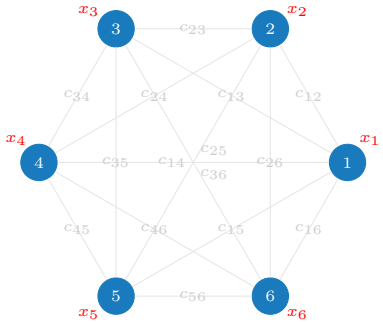
Covariance Data Relationships

- Data sample $\mathbf{x} = [x_1, \ldots, x_N]^\mathsf{T}$ with covariance $\mathbf{C}$
  - $\Rightarrow$ Build a **graph** where:
  - $\Rightarrow$ the features are the node signals x_i
  - $\Rightarrow$ the edges are the covariance values c_{ij} $\rightarrow$ fully-connected graph



Covariance Filters

- Definition: Graph convolution covariance filters

$$\mathbf{z} = \mathbf{H}(\hat{\mathbf{C}})\mathbf{x} = \sum_{k=0}^{K} h_k \hat{\mathbf{C}}^k \mathbf{x}$$

$\Rightarrow$ learnable parameters: h_k

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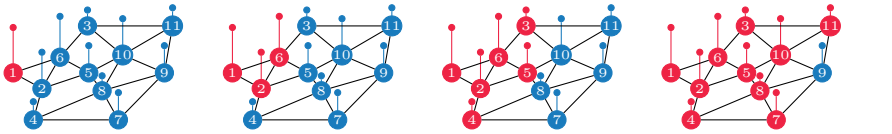
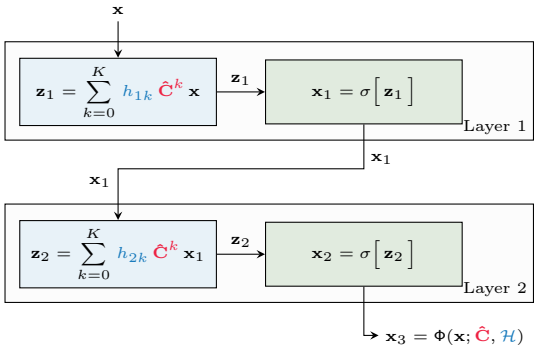


- $\hat{\mathbf{C}}^k\mathbf{x}$ shifts signal $\mathbf{x}$ k times over the covariance graph $\hat{\mathbf{C}}$

Covariance Neural Networks (VNNs)

- Definition: Covariance filters followed by pointwise nonlinearities σ

$$\mathbf{x}^l = \sigma\left(\mathbf{H}^l(\hat{\mathbf{C}})\mathbf{x}^{l-1}\right) \quad l = 1, \ldots, L.$$



Sihag, Mateos, McMillan & Ribeiro. coVariance Neural Networks. NeurIPS, 2022

Connections to PCA

- The covariance filter matrix has the form

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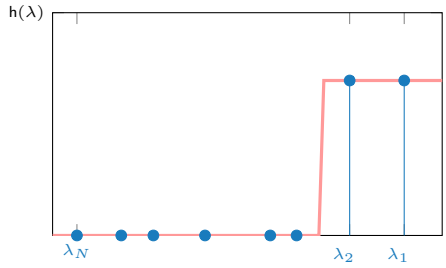
which means that in the spectrum it acts as a polynomial

$$h(\hat{\lambda}_i) = \sum_{k=0}^{K} h_k \hat{\lambda}_i{}^k$$

Connections to PCA

- PCA does a hard filtration by selecting only a few principal directions



PCA frequency response

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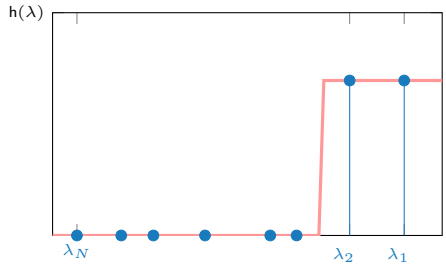

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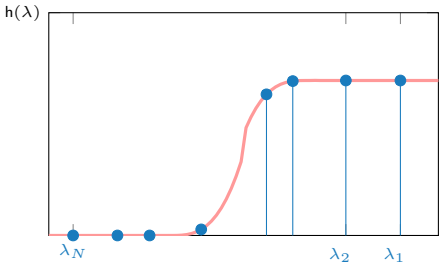

VNN frequency response

- VNNs learn to modulate each principal direction to solve the task

Stability of VNNs

- We train the VNN on the noisy covariance matrix $\hat{\mathbf{C}}$

  $\Rightarrow$ How does this affect the performance?

- Stability for true $\mathbf{H}(\mathbf{C})$ vs. estimated $\mathbf{H}(\hat{\mathbf{C}})$ covariance filter:

$$\|\mathbf{H}(\hat{\mathbf{C}}) - \mathbf{H}(\mathbf{C})\| \leq \frac{P}{\sqrt{t}} \mathcal{O}\left(\sqrt{N} + \frac{\|\mathbf{C}\| \sqrt{\log(Nt)}}{\nu t}\right)$$

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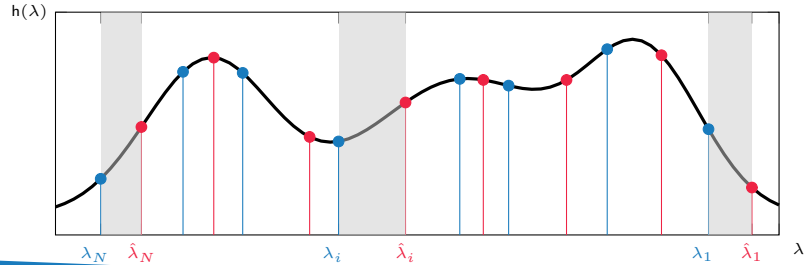
- P : Lipschitz constant of the filter – variability of the frequency response



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- P : Lipschitz constant of the filter – variability of the frequency response
- t and N: number of samples and data dimension
- Data distribution

Stability of VNNs

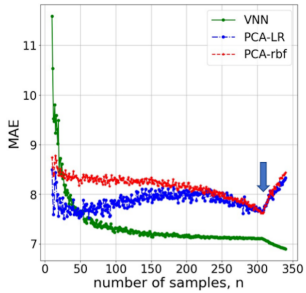
- Stability bound for covariance neural network:

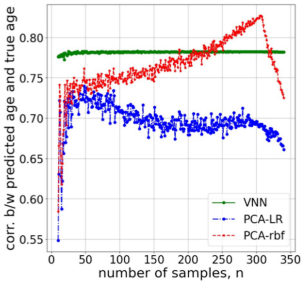
$$\|\mathbf{H}(\hat{\mathbf{C}})-\mathbf{H}(\mathbf{C})\| \leq \frac{P}{\sqrt{t}}\mathcal{O}\left(\sqrt{N}+\frac{\|\mathbf{C}\|\sqrt{\log(Nt)}}{\nu t}\right)=\beta$$

$$\|\mathbf{\Phi}(\mathbf{x},\hat{\mathbf{C}},\mathcal{H})-\mathbf{\Phi}(\mathbf{x},\mathbf{C},\mathcal{H})\| \leq LF^{L-1}\beta$$

  - $\Rightarrow$ Bound for the neural network depends on the filter bound
  - $\Rightarrow$ Increases with the number of layers L and filter bank size F
  - $\Rightarrow$ The non-linearity spreads the energy across the spectrum

- Setup:
  - ⇒ Train with one covariance, test with another with different number of samples
  - ⇒ Brain dataset: predict brain age
  - ⇒ VNN vs PCA+SVM





- VNN performance is more consistent

Limitations of VNNs

- So are VNNs enough?

  $\Rightarrow$ No

- Limitations

  $\Rightarrow$ Low-data high-dimensional settings $\rightarrow$ sample covariance estimator is really bad!

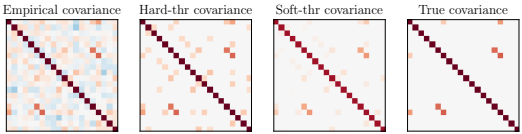
  $\Rightarrow$ Computationally inefficient $\rightarrow$ complexity $\mathcal{O}(N^2)$

Outline

- Covariance Neural Networks
- **Sparse VNNs**
- Spatiotemporal VNNs
- Fair VNNs
- Conclusions

Sparse Covariance Matrices

- In practice, the true covariance matrix $\mathbf{C}$ is sparse



- Hard thresholding:

$$[\bar{\mathbf{C}}]_{ij} = \begin{cases} \hat{c}_{ij} & \text{if } |\hat{c}_{ij}| \geq \tau/\sqrt{t} \\ 0 & \text{otherwise} \end{cases}$$

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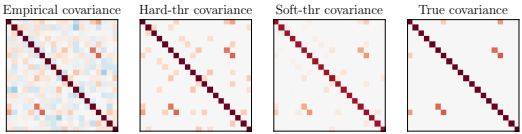

- Hard thresholding:

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- Soft thresholding:

$$[\bar{\mathbf{C}}]_{ij} = \begin{cases} \hat{c}_{ij} - \text{sign}(\hat{c}_{ij})\tau/\sqrt{t} & \text{if } |\hat{c}_{ij}| \geq \tau/\sqrt{t} \\ 0 & \text{otherwise} \end{cases}$$

Sparse Covariance Neural Networks (S-VNNs)

- Stability for true $\mathbf{H}(\mathbf{C})$ vs. hard-thresholded $\mathbf{H}(\bar{\mathbf{C}})$ covariance filter:

$$\|\mathbf{H}(\bar{\mathbf{C}}) - \mathbf{H}(\mathbf{C})\| \leq \frac{1}{\sqrt{t}} P c_0 \sqrt{N \log N}(1 + \sqrt{2N}) + \mathcal{O}\left(\frac{1}{t}\right).$$

- c_0: number of non-zero elements in $\bar{\mathbf{C}}$

  $\Rightarrow$ $c_0 \ll \|\mathbf{C}\| \rightarrow$ improved stability!

  $\Rightarrow$ We need fewer data

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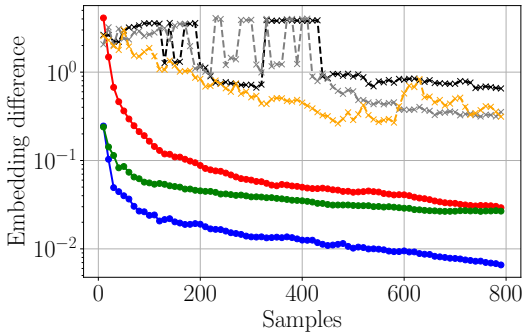

Sparse Covariance Neural Networks (S-VNNs)

- Stability for true $\mathbf{H}(\mathbf{C})$ vs. soft-thresholded $\mathbf{H}(\bar{\mathbf{C}})$ covariance filter:

$$\|\mathbf{H}(\bar{\mathbf{C}}) - \mathbf{H}(\mathbf{C})\| \leq \frac{1}{\sqrt{t}} P\sqrt{N} C c_0 \max(1,\lambda_{max}) \sqrt{\max\left(\log(N/c_0^2), 1\right)} (1 + \sqrt{2N}) + \mathcal{O}\left(\frac{1}{t}\right).$$

- $\sqrt{\max\left(\log(N/c_0^2), 1\right)} \leq \sqrt{\log N} \rightarrow$ further improved stability!

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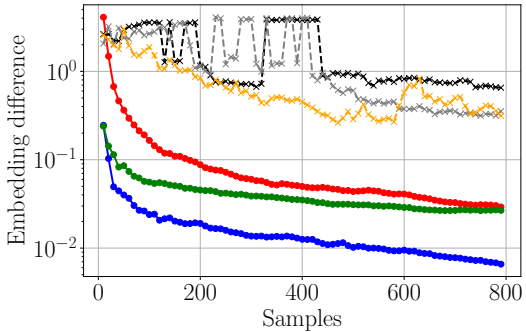
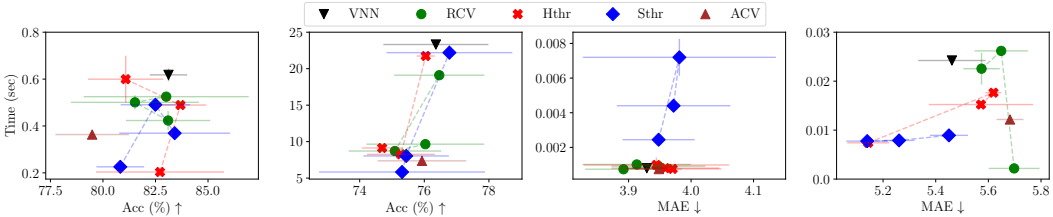
- $\sqrt{\max\left(\log(N/c_0^2), 1\right)} \leq \sqrt{1}$



Numerical Results

- Brain recordings: classify patient condition
- Human Action Recognition: classify action performed



Datasets: MHEALTH, Realdisp, ADNI1 and ADNI2

- S-VNNs are always faster and more accurate due to spurious correlation removal

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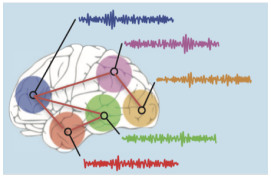
VNNs and Temporal Data

- So far:
  - ⇒ VNNs and variants work well on static and batch data
  - ⇒ Yet, data are often time-varying and non-stationary



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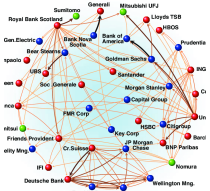

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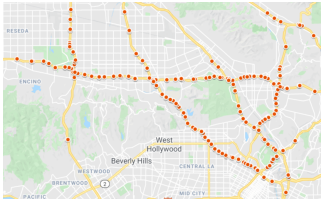

Traffic

- Goal: Incorporate spatiotemporal interdependencies

SpatioTemporal Covariance Neural Networks (STVNNs)

- Online covariance estimate

$$\hat{\mathbf{C}}_{t+1} = \xi_t \hat{\mathbf{C}}_t + \zeta_t \mathbf{x}_{t+1} \mathbf{x}_{t+1}^{\mathsf{T}}$$

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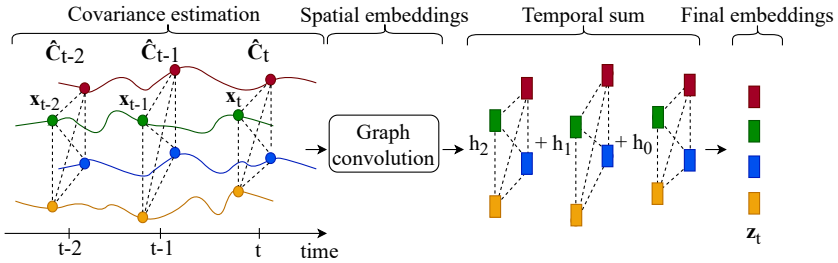
- SpatioTemporal coVariance Filter (STVF)

$$\mathbf{z}_t := \mathbf{H}(\hat{\mathbf{C}}_t, \mathbf{h}_t, \mathbf{x}_{T:t}) = \sum_{t'=0}^{T-1} \sum_{k=0}^{K} h_{kt'} \hat{\mathbf{C}}_t^k \mathbf{x}_{t-t'}$$

$\Rightarrow$ Spatial (graph) and temporal convolutions



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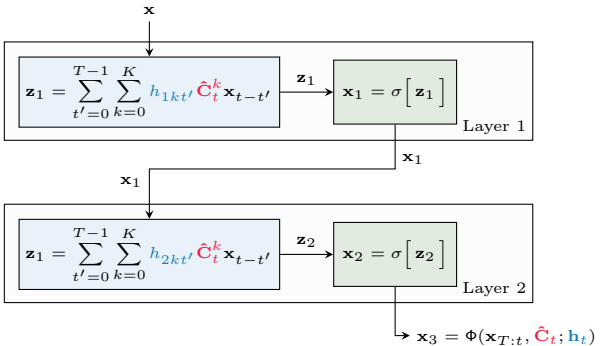
- SpatioTemporal coVariance Neural Network (STVNN):

  $\Rightarrow$ Sequence of filterbanks followed by non-linearity

$$\mathbf{z}_t^l = \sigma\left(\mathbf{H}^l(\hat{\mathbf{C}}_t, \mathbf{h}_t, \mathbf{z}_{T:t}^{l-1})\right)$$



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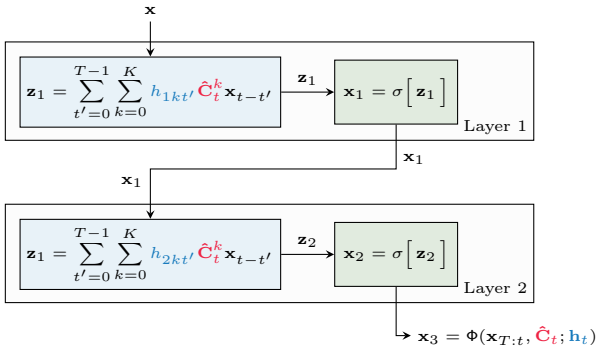

- Online parameter updates

$$\mathbf{h}_{t+1} = \mathbf{h}_t - \eta \nabla_t \mathcal{L}(\mathbf{\Phi}(\mathbf{x}_{T:t}, \hat{\mathbf{C}}_t; \mathbf{h}_t))$$

Stability of STVNNs

- Stability for true $\mathbf{H}(\mathbf{C}, \mathbf{h}^*, \mathbf{x}_{T:t})$ vs. estimated $\mathbf{H}(\hat{\mathbf{C}}_t, \mathbf{h}_t, \mathbf{x}_{T:t})$ spatiotemporal online covariance filter:

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Cavallo, Sabbaqi & Isufi. Spatiotemporal Covariance Neural Networks. ECML PKDD, 2024

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- STVNNs are stable to both the covariance and parameter estimation errors

Forecasting Performance

- **Baselines**
  - ⇒ TPCA: temporal PCA
  - ⇒ VNN: VNN with temporal data as features
  - ⇒ VNNL: VNN on each snapshot followed by LSTM for sequence modeling

Datasets	Steps	LSTM	TPCA	VNN	VNNL	STVNN
NOAA	1	1.98	2.42	1.71	*1.67*	**1.35**
	3	**3.10**	3.93	*3.14*	3.36	3.21
	5	**3.46**	5.10	4.03	5.76	*3.71*
Molene	1	0.29	0.35	**0.20**	0.21	**0.20**
	3	0.47	0.46	0.43	*0.41*	**0.38**
	5	0.60	**0.56**	0.64	0.59	**0.56**
Exchange rate	1	1.25	1.73	0.70	*0.68*	**0.65**
	3	1.33	1.66	*0.98*	1.00	**0.94**
	5	1.39	1.75	1.19	*1.16*	**1.11**

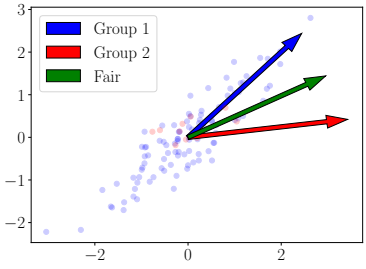
- STVNN more consistent across tasks and datasets
- STVNN better than TPCA for stability and than VNN and VNNL for temporal convolution

Outline

- Covariance Neural Networks
- Sparse VNNs
- Spatiotemporal VNNs
- **Fair VNNs**
- Conclusions

Limited Data as Source of Unfairness

- Datasets might contain harmful biases:
  - ⇒ E.g., underrepresented groups
  - ⇒ VNNs might have unfair performance
  - ⇒ Fair PCA might fail due to instability
- Solution: Fair-VNNs!
  - ⇒ How can VNN's stability help?
  - ⇒ How to make VNNs fair?



Fair Covariance Neural Networks (F-VNNs)

- Definition: *equality of performance across groups*

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$$\mathbf{C}_{bal} = \alpha\hat{\mathbf{C}} + (1-\alpha)(\hat{\mathbf{C}}_h - \hat{\mathbf{C}}_g) = \alpha_g\hat{\mathbf{C}}_g + \alpha_h\hat{\mathbf{C}}_h$$

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- Bias mitigation penalties

  $\Rightarrow$ F-VNNs are trained with a loss penalty to encourage fairness

$$\min_{\mathcal{H}} \ \gamma\mathcal{L}(\mathbf{X}, \mathbf{y}, \Phi) + (1-\gamma)\mathcal{R}(\mathbf{X}, \mathbf{y}, \mathbf{z}, \Phi)$$

  $\Rightarrow$ $\mathcal{L}$ is task-specific loss (e.g., cross-entropy)

  $\Rightarrow$ $\mathcal{R}$ is bias penalty (e.g., performance difference across groups)

  $\Rightarrow$ γ is balancing term

Stability of F-VNNs

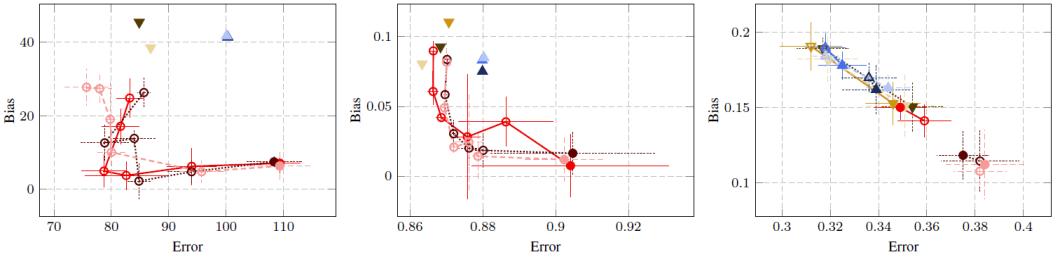
- Fair covariance estimates
  - $\Rightarrow$ $\mathbf{C}_{bal}$ is subject to covariance estimation errors
  - $\Rightarrow$ Fair PCA is unstable $\rightarrow$ might result in biased treatment
- VNNs' stability as fairness
  - $\Rightarrow$ F-VNNs are stable with fair covariance estimates:

$$\|\mathbf{H}(\mathbf{C}_{bal}) - \mathbf{H}(\mathbf{C})\| \leq P\sqrt{N+2N^2}\left(\mathcal{O}\left(\frac{1}{\sqrt{T_g}}\right) + \mathcal{O}\left(\frac{1}{\sqrt{T_h}}\right)\right)$$

  - $\Rightarrow$ Stability corresponds to better fairness as it guarantees consistent behavior across groups

Numerical Results

Dataset	Description	Task	Sensitive attribute
Parkinson (left)	Medical records of patients	Regression for Parkinson's level	Gender
LSAC (center)	Law school students' features	Regression for GPA	Race
German credit (right)	Features of individuals applying for credit	Classification (good or bad)	Gender



- F-VNNs are flexible in the bias vs error tradeoff by changing γ
- Generally, F-VNNs perform better than PCA and improve fairness

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- Thank you for the attention! **Main references and code:**
  - ⇒ Sihag, Mateos, McMillan & Ribeiro. coVariance Neural Networks. NeurIPS, 2022
  - ⇒ Cavallo, Sabbaqi & I. Spatiotemporal Covariance Neural Networks. ECML PKDD, 2024
  - ⇒ Cavallo, Navarro, Segarra & I. Fair Covariance Neural Networks. under review, 2024